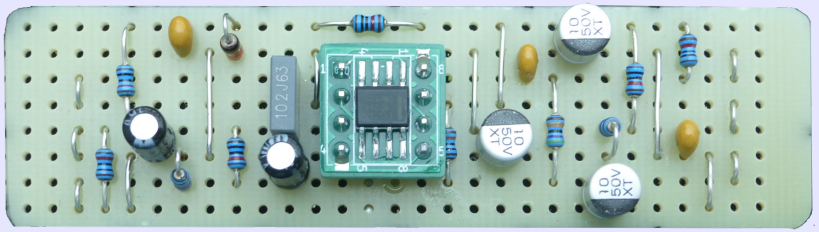
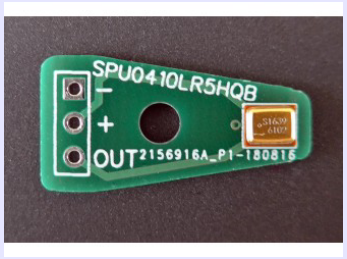
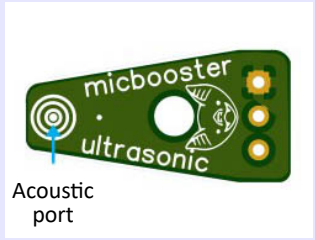
MEMS: 48v Phantom Powered Ultrasonic Mic Interface

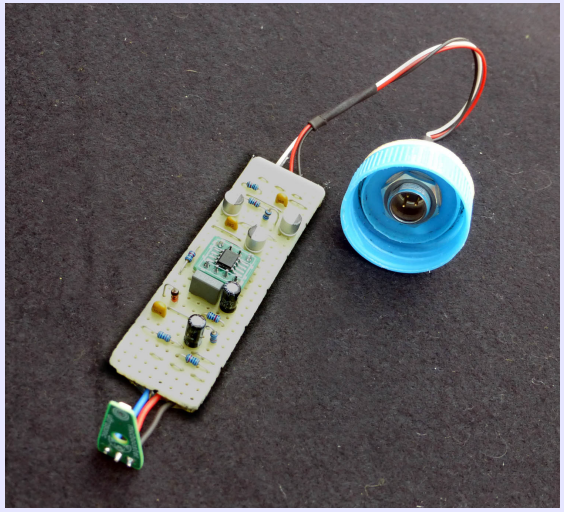
- Interfaces a MEMS capsule to a 48v phantom powered mic pre-amp
- Designed to amplify ultrasonic bat echolocation calls
- Simple stripboard layout





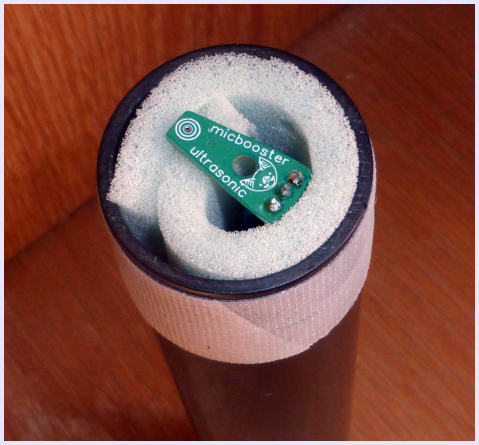


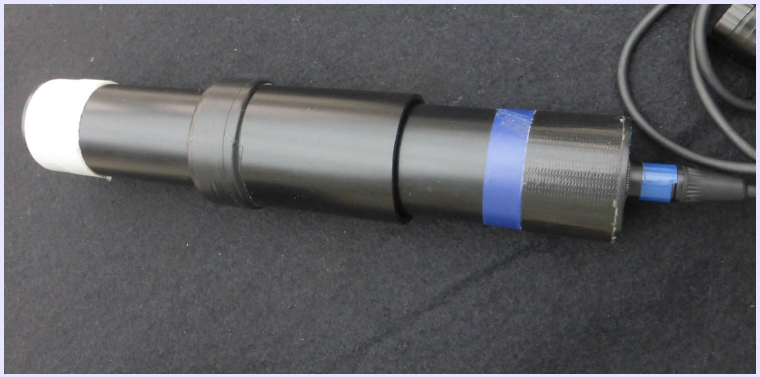
Mic Capsule: https://micbooster.com/product/ultrasonic-mic-board



Mini XLR holder: 32mm dia. bottle cap or similar





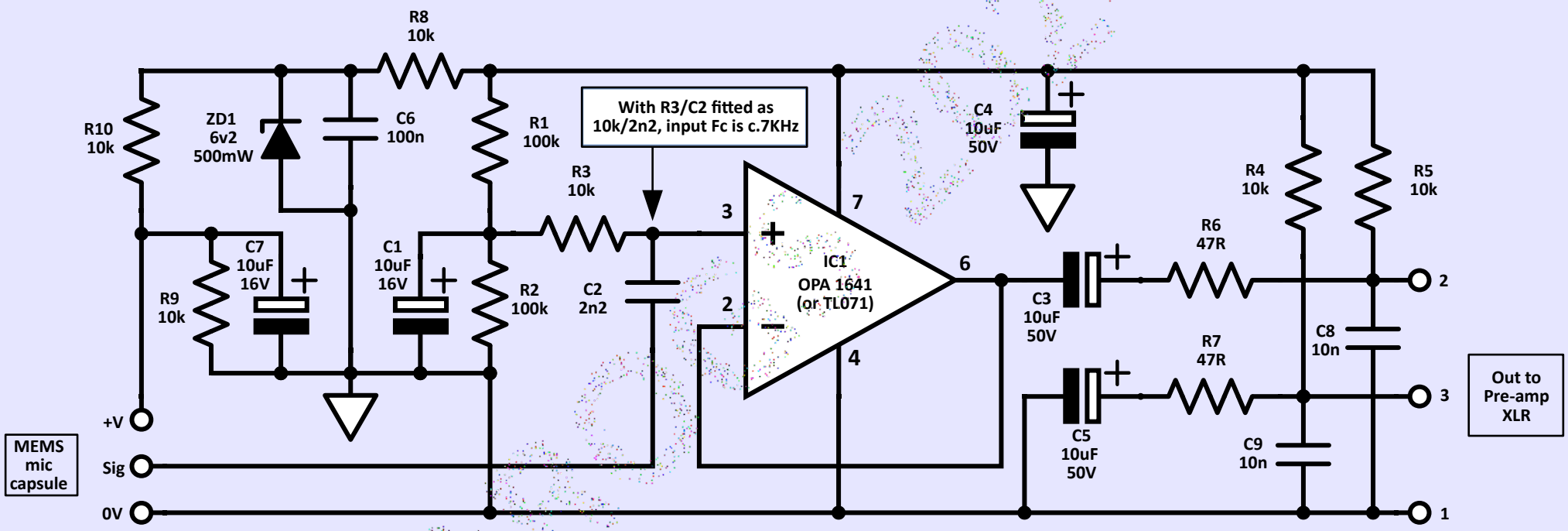


Mic Housing: 200mm x 32mm dia. Polypropylene pipe



https://zoom-ultrasonic.github.io/ - © rogs 11.26

• MEMS 48v Phantom Powered Ultrasonic Mic Pre-amp - SCHEMATIC •



Notes:

- All input and output connectors are 'doubled padded', to enable more reliable terminations to be made onto the track side of the stripboard.
- Note that the order of both the input & output terminations are different on the stripboard.

© rogs 9.24

MEMS 48v Ultrasonic Microphone Pre-amp

Overview:

The circuitry shown in the schematic is designed to interface a Knowles MEMS microphone capsule - type SPU0410LR5H - with a conventional balanced microphone pre-amplifier, equipped with 48v phantom power, and a 3 pin XLR connection.

The circuit is specifically designed to record ultrasonic 'echolocation' signals emitted by bats, and as such does not have a conventional audio range frequency response.

The strip-board layout that the components are fitted to is intended to be mounted into a lightweight enclosure – typically a short length of 32mm diameter plastic waste pipe. This will allow for the creation of a lightweight microphone enclosure, suitable for mounting onto a boom pole.

Circuit description:

The active device used for this project can be either the Texas Instruments OPA1641 operational amplifier (op-amp) for best performance, or a TL071 for simpler construction. Both allow a low noise, low distortion, impedance converting amplifier to be constructed. (The OPA1641 has slightly better figures, but does require a DIL to SOIC adaptor).

The op-amp is configured in this project as a non-inverting, unity gain buffer amplifier. This is achieved by connecting the amplifier output directly to the inverting input.

The amplifier is powered from the 48v phantom power of the attached microphone pre-amp. Resistors R4 and R5 are connected to pins 3 and 2 of the XLR input connector. The other ends of both resistors are joined and - together with the decoupling capacitor C4 - allow a DC voltage of c.24v to be applied to pin 7 of the op-amp.

The actual DC value is not critical, and the values of R4 and R5 are selected to ensure the maximum permitted supply to the op-amp cannot be exceeded.

The op-amp requires a 'dual' voltage supply, and this is provided by the creation of a 'half rail' voltage of around 12v by the voltage divider R1 and R2, together with the decoupling capacitor C1.

This 'half rail' voltage is required to bias the op-amp, so that the output can swing symmetrically around this reference voltage. It is fed to the non-inverting input of the op-amp via R3.

The signal input from the microphone capsule itself is fed to the non-inverting input of the op-amp via C2. The values of R3 and C2 are selected to provide a high pass filtered input signal, where the 'cut off' (-3dB) frequency is set at c.7KHz. This will permit the lowest frequency of bat echolocation signals to be recorded, but will attenuate the vast majority of 'normal' audio frequency signals, which only serve to confuse with this type of recording.

The MEMS microphone capsule requires a separate DC supply to be applied to the **+v** terminal on the schematic. This DC should have a value within the range 1.5 and 3v, and in this case the value is selected to be c.2.5v.

A 6v2 Zener diode is fed with DC from the R4/R5 summing junction via R8. This point is de-coupled by C6 to remove any Zener noise.

A second voltage divider formed by R10 and R9 - decoupled by C7 – would normally create an unloaded voltage of c.3.1v at the junction of the two resistors, but R10 additionally supplies extra current to the capsule itself, and so the on load voltage at that point is actually c.2.5v.
The Zener is included to protect the MEMs capsule, which can be sensitive to any over voltage.

The op-amp ultrasonic audio output is taken from pin 6 and fed, via C3 and R6, to pin 2 (hot) of the XLR connector. C3 is included to isolate the op-amp DC level from the phantom power present on the XLR connector.

Pin 3 (cold) of the XLR connector is connected to AC ground via R7 and C5. There is no audio on this pin, and those components merely serve to allow for an impedance balanced output to be created, to help optimise the CMRR and minimise any noise pick up in the connecting cable.

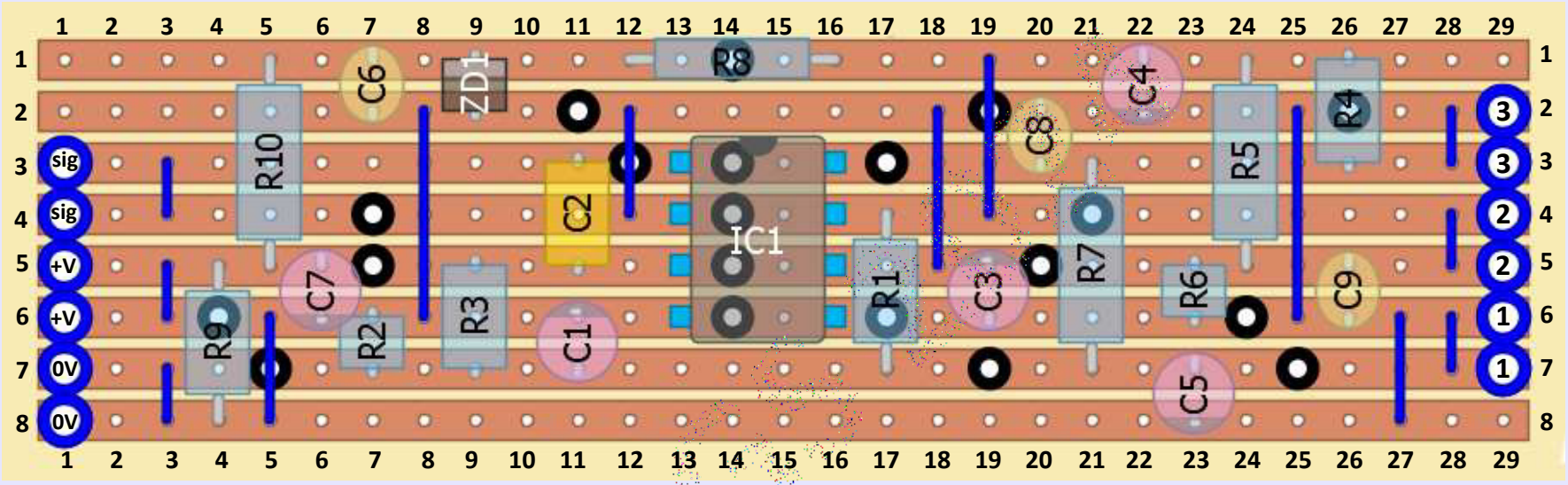
C8 an C9 are fitted to help remove any stray RF interference that may be present in the vicinity of microphone operation.

https://zoom-ultrasonic.github.io/ rogs 11.26

MEMS 48v Phantom Powered Ultrasonic Mic Pre-amp - Stripboard layout

© rogs 10.24

From MEMs Capsule



To Mic pre-amp (Via Mini XLR)

- 20 x 'spin off' track cuts at black and white dot locations
- 13 x TCW links at blue line locations (25 gauge or similar)
- *The 'double' termination reference numbers allow for external wire connections to be made between tracks – on the track side – for a more robust and reliable connection.*

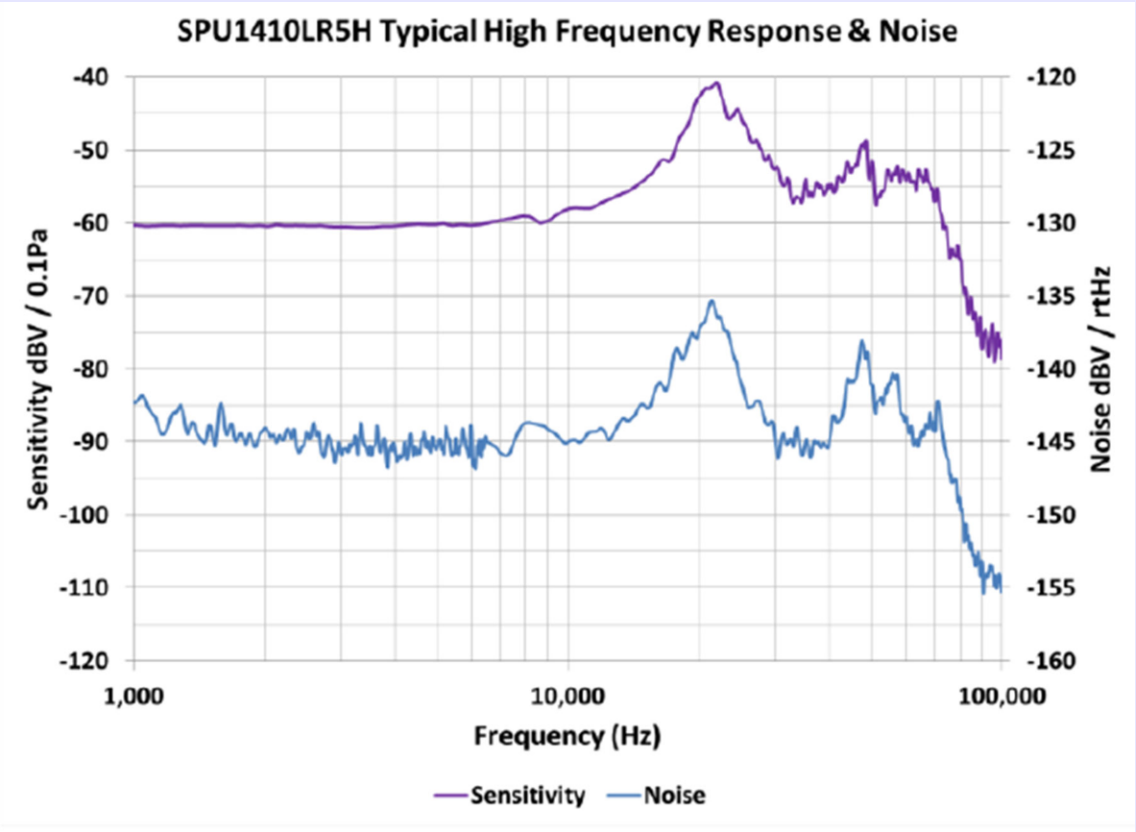
Parts List

- Capacitors:
- *'elec' = electrolytic*
- *'mlc' = multi layer ceramic*
- *'PET' = Polyethylene film*
- C1 & 7 are 10uF 16v elec
- C2 is 2.2nF 63v PET
- C3,4 & 5 are 10uF 50v elec
- C6 is 100nF 63v mlc
- C8 & C9 are 10nF 63v mlc

- *All resistors are MF12 1/8th watt metal film:*
- R1,R2 are 100k
- R3,4,5,8,9 & 10 are 10k
- R6,R7 are 47R
- IC1 OPA1641 (Pin1 is column 13, row 3)
- Use SOIC DIP adaptor for IC1

(TL071 can be used as a DIP equivalent op-amp, but has a slightly higher noise level)

- ZD1 6v2 Zener diode 500mW



The graph above shows the HF response of the Knowles SPU1410LR5H MEMs microphone capsule…..

As you can see, it's not very linear! .. An erratic response like this is difficult to correct in the analogue mode - in real time - without complex filters being applied.

However, processing recorded files in the digital mode, by the sequential application of the 2 equalisation processes described below, can make frequency response correction a simple procedure.

The detailed notes below describe a script used in Adobe Audition to perform the task on the 32 bit float files recorded with a Zoom F3 recorder.
(Other DAW/Recorder combinations should offer similar functions).

Adobe Audition Script for SPU1410 MEMs mic HF frequency response correction

• **'BATX LF cut off' Graphic EQ preset**: *30 band settings (Range: 120dB)....* ***All set to -60dB except****:*

• 10KHz • 12.5KHz • 16KHz • 20Khz • 25KHz - *Those 5 bands are set at 0dB*

• **'1410' Parametric EQ preset**:

• *Range setting: 10kHz to 85KHz* • *Constant Q* • *Gain: 6dB*

4 Band settings (5th band not used):

• 25KHz: (Q 5) -15dB • 45KHz: (Q5) +3dB • 65KHz:(Q 5) -3dB • 80KHz: (Q 10) +9dB

Script: *The following actions are applied in sequence:*

• Apply **BATX LF cut off**
• Apply **1410** Parametric EQ filter
• Convert 32 bit float file to 16 bit
• End of Script

© rogs 06.25